Classification of Pets – PROJECT

Riya Harugop


```
labels
```

```
['Abyssinian',
 'Abyssinian',
 'Abyssinian',
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Code JupyterLab Python [conda env:base] * Anaconda Toolbox

[10]:
```
!pip install tensorflow
import tensorflow as tf
from tensorflow.keras.preprocessing.image import load_img, img_to_array
```

```
^C
WARNING: Connection timed out while downloading.
WARNING: Attempting to resume incomplete download (0 bytes/212 kB, attempt 1)
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the followin
g dependency conflicts.
streamlit 1.51.0 requires protobuf<7,>=3.20, but you have protobuf 7.35.0 which is incompatible.
Collecting tensorflow
  Using cached tensorflow-2.21.0-cp313-cp313-win_amd64.whl.metadata (4.5 kB)
Collecting absl-py>=1.0.0 (from tensorflow)
  Using cached absl_py-2.4.0-py3-none-any.whl.metadata (3.3 kB)
Collecting astunparse>=1.6.0 (from tensorflow)
  Using cached astunparse-1.6.3-py2.py3-none-any.whl.metadata (4.4 kB)
Collecting flatbuffers>=25.9.23 (from tensorflow)
  Using cached flatbuffers-25.12.19-py2.py3-none-any.whl.metadata (1.0 kB)
Collecting gast!=0.5.0,!=0.5.1,!=0.5.2,>=0.2.1 (from tensorflow)
  Using cached gast-0.7.0-py3-none-any.whl.metadata (1.5 kB)
Collecting google_pasta>=0.1.1 (from tensorflow)
  Using cached google_pasta-0.2.0-py3-none-any.whl.metadata (814 bytes)
Collecting libclang>=13.0.0 (from tensorflow)
  Using cached libclang-18.1.1-py2.py3-none-win_amd64.whl.metadata (5.3 kB)
Collecting opt_einsum>=2.3.2 (from tensorflow)
  Using cached opt_einsum-3.4.0-py3-none-any.whl.metadata (6.3 kB)
Requirement already satisfied: packaging in c:\users\riyah\anaconda3\lib\site-packages (from tensorflow) (25.0)
Collecting protobuf<8.0.0,>=6.31.1 (from tensorflow)
```

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Code JupyterLab Python [conda env:base] * Anaconda Toolbox

```
     ---------------------------------------- 8/16 [h5py]
     ---------------------------------------- 9/16 [grpcio]
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     ---------------------------------------- 10/16 [google_pasta]
     ---------------------------------------- 10/16 [google_pasta]
     ---------------------------------------- 10/16 [google_pasta]
     ---------------------------------------- 11/16 [gast]
     ---------------------------------------- 13/16 [absl-py]
     ---------------------------------------- 13/16 [absl-py]
     ---------------------------------------- 13/16 [absl-py]
     ---------------------------------------- 13/16 [absl-py]
     ---------------------------------------- 14/16 [keras]
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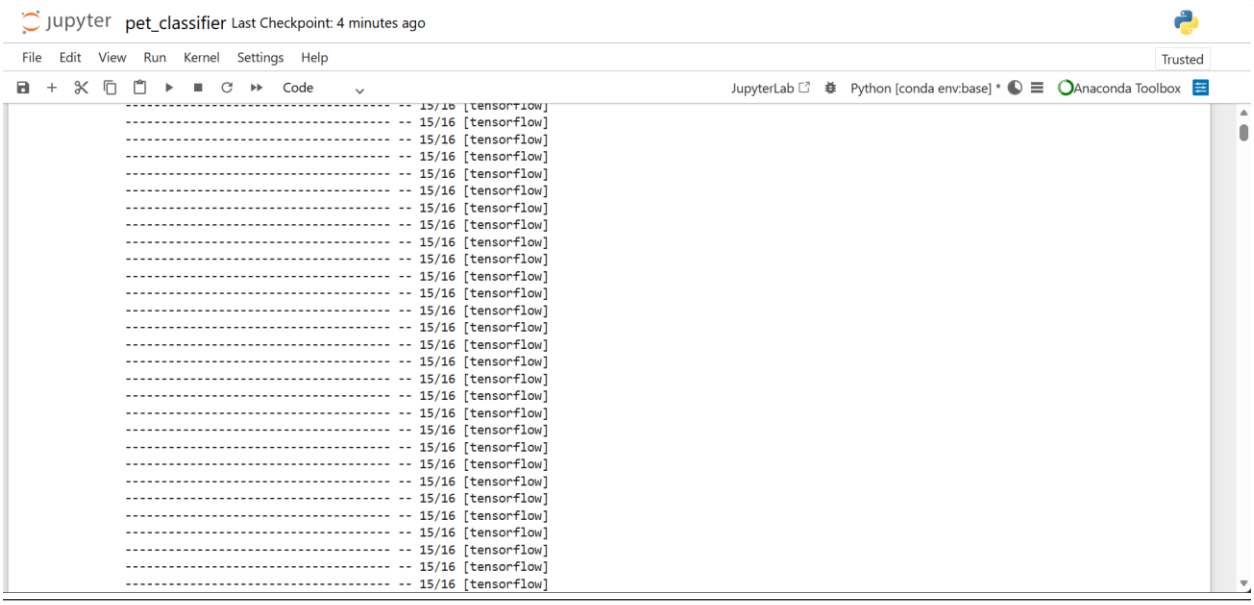
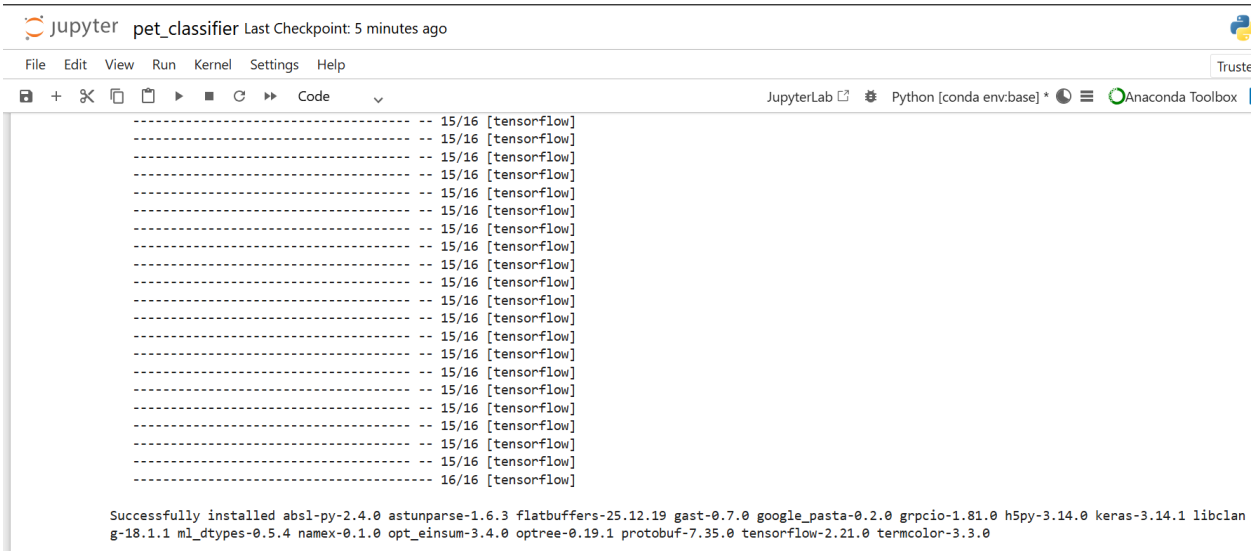
     ---------------------------------------- 14/16 [keras]
     ---------------------------------------- 14/16 [keras]
```






```
Successfully installed absl-py-2.4.0 astunparse-1.6.3 flatbuffers-25.12.19 gast-0.7.0 google_pasta-0.2.0 grpcio-1.81.0 h5py-3.14.0 keras-3.14.1 libclan
g-18.1.1 ml_dtypes-0.5.4 namex-0.1.0 opt_einsum-3.4.0 optree-0.19.1 protobuf-7.35.0 tensorflow-2.21.0 termcolor-3.3.0
```

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Code JupyterLab Python [conda env:base] *

[20]:
```
features = []
labels = []

IMAGE_SIZE = (224, 224)

for i, name in enumerate(image_names):

    label = ' '.join(name.split('_')[:-1])

    label_encoded = label_to_id[label]

    img = load_img(os.path.join(images_fp, name))

    img = tf.image.resize_with_pad(
        img_to_array(img, dtype='uint8'),
        *IMAGE_SIZE
    ).numpy().astype('uint8')

    features.append(img)
    labels.append(label_encoded)

    if i % 500 == 0:
        print(f"Processed {i}/{len(image_names)}")

print("Done")
```

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Code JupyterLab Python [conda env:base] * Anaconda Toolbox

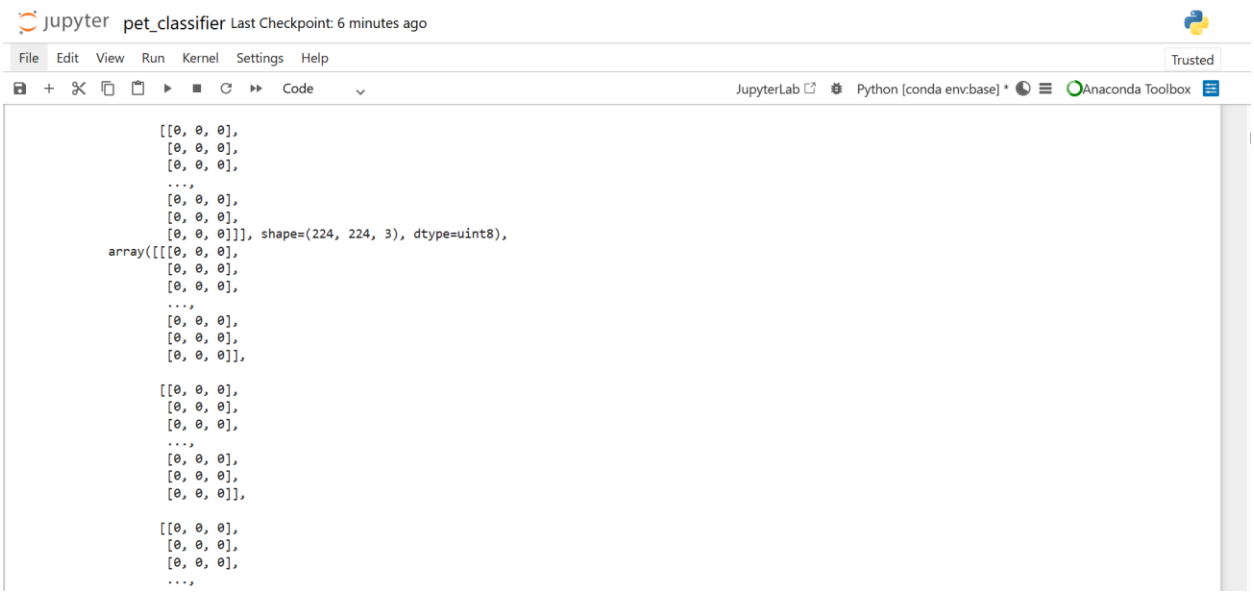
```
Processed 500/7390
Processed 1000/7390
Processed 1500/7390
Processed 2000/7390
Processed 2500/7390
Processed 3000/7390
Processed 3500/7390
Processed 4000/7390
Processed 4500/7390
Processed 5000/7390
Processed 5500/7390
Processed 6000/7390
Processed 6500/7390
Processed 7000/7390
Done
```

[21]:
```
print(len(features))
print(len(labels))
```

```
7390
7390
```

[22]:
```
features
```

[22]:
```
[array([[[0, 0, 0],
         [0, 0, 0],
         [0, 0, 0],
         ...,
         [0, 0, 0],
```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Code
JupyterLab
Python [conda env:base] *
Anaconda Toolbox
[[0, 0, 0],
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[0, 0, 0],
...,
[0, 0, 0],
[0, 0, 0],
[0, 0, 0]]], shape=(224, 224, 3), dtype=uint8),
array([[[0, 0, 0],
[0, 0, 0],
[0, 0, 0],
...,
[0, 0, 0],
[0, 0, 0],
[0, 0, 0]],
[[0, 0, 0],
[0, 0, 0],
[0, 0, 0],
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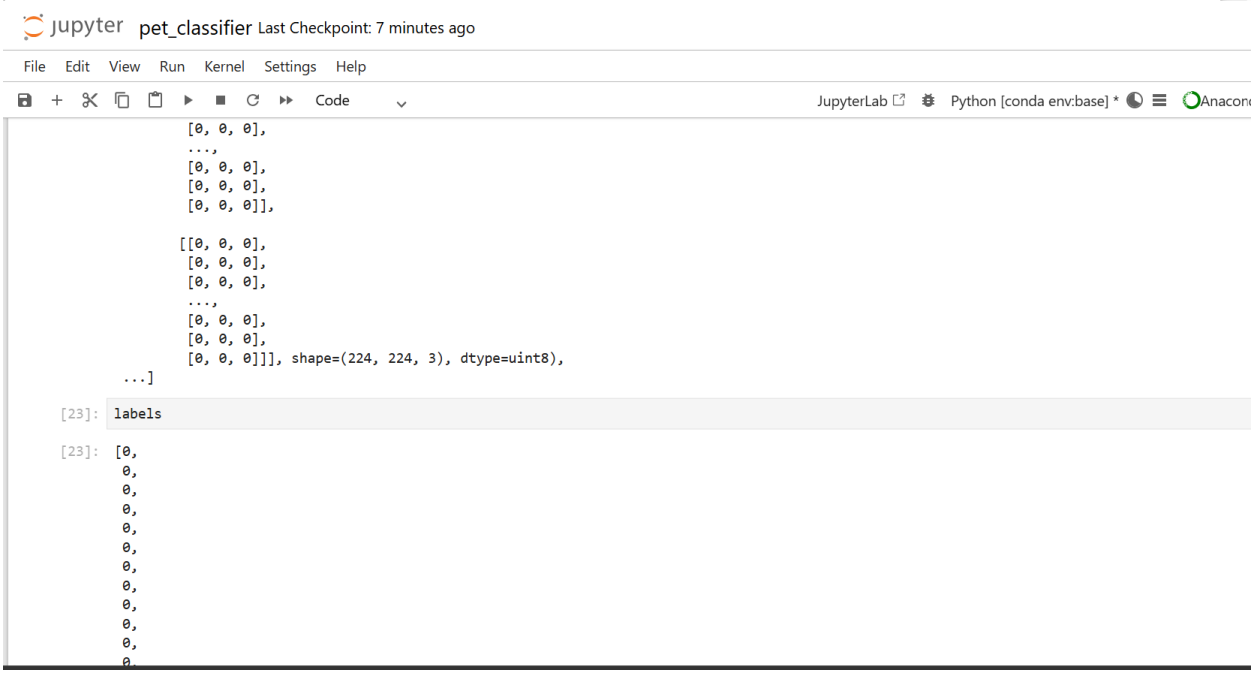
[0, 0, 0],
[0, 0, 0],
[0, 0, 0]],
[[0, 0, 0],
[0, 0, 0],
[0, 0, 0],
...,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Code
JupyterLab
Python [conda env:base] *
[0, 0, 0],
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[0, 0, 0],
[0, 0, 0]],
[[0, 0, 0],
[0, 0, 0],
[0, 0, 0],
...,
[0, 0, 0],
[0, 0, 0],
[0, 0, 0]]], shape=(224, 224, 3), dtype=uint8),
...]
[23]: labels
[23]: [0,
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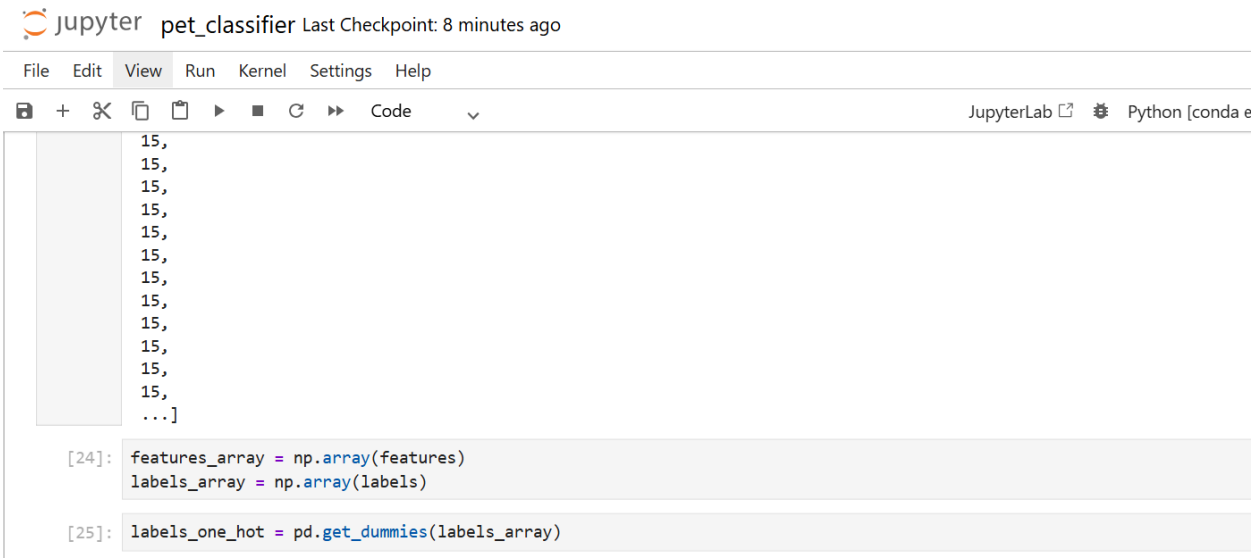

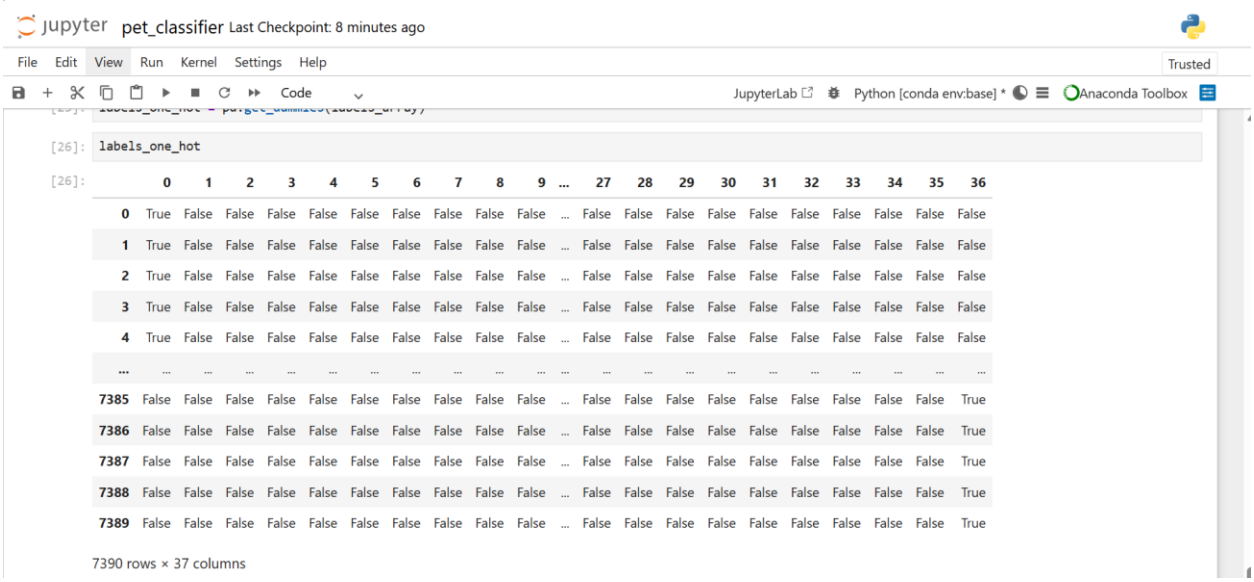

	0	1	2	3	4	5	6	7	8	9	...	27	28	29	30	31	32	33	34	35	36
0	True	False	False	False	False	False	False	False	False	False	...	False	False	False	False	False	False	False	False	False	False
1	True	False	False	False	False	False	False	False	False	False	...	False	False	False	False	False	False	False	False	False	False
2	True	False	False	False	False	False	False	False	False	False	...	False	False	False	False	False	False	False	False	False	False
3	True	False	False	False	False	False	False	False	False	False	...	False	False	False	False	False	False	False	False	False	False
4	True	False	False	False	False	False	False	False	False	False	...	False	False	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
7385	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False	False	False	False	False	False	True
7386	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False	False	False	False	False	False	True
7387	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False	False	False	False	False	False	True
7388	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False	False	False	False	False	False	True
7389	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False	False	False	False	False	False	True

7390 rows × 37 columns

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Code JupyterLab Python [conda env:base] * Anaconda Toolbox

[27]:
```
from sklearn.model_selection import train_test_split
```

[32]:
```
X_train, X_test, y_train, y_test = train_test_split(
    features_array,
    labels_one_hot.values,
    test_size=0.2,
    random_state=42
)
```

[33]:
```
X_train, X_val, y_train, y_val = train_test_split(
    X_train,
    y_train,
    test_size=0.25,
    random_state=1
)
```

[34]:
```
from tensorflow.keras import layers, Input, Model
from tensorflow.keras.models import Sequential
from tensorflow.keras.applications import ResNet50
from tensorflow.keras.applications.resnet50 import preprocess_input as pp_i
from tensorflow.keras.layers import RandomFlip, RandomRotation, Dense, Dropout
from tensorflow.keras.losses import CategoricalCrossentropy
from tensorflow.keras.optimizers import Adam
```

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Code JupyterLab Python [conda env:base] * Anaconda Toolbox

[36]:
```
data_augmentation = Sequential([
    RandomFlip("horizontal_and_vertical"),
    RandomRotation(0.2)
])

prediction_layer = Dense(37, activation='softmax')
```

[37]:
```
resnet_model = ResNet50(
    include_top=False,
    pooling='avg',
    weights='imagenet'
)

resnet_model.trainable = False
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/resnet/resnet50_weights_tf_dim_ordering_tf_kernels_notop.h5
94765736/94765736 ━━━━━━━━━━━━━━━━━━━━ 24s 0us/step
```

[38]:
```
# Build Model

inputs = Input(shape=(224, 224, 3))

x = data_augmentation(inputs)
x = pp_i(x)
x = resnet_model(x, training=False)
x = Dropout(0.2)(x)
```


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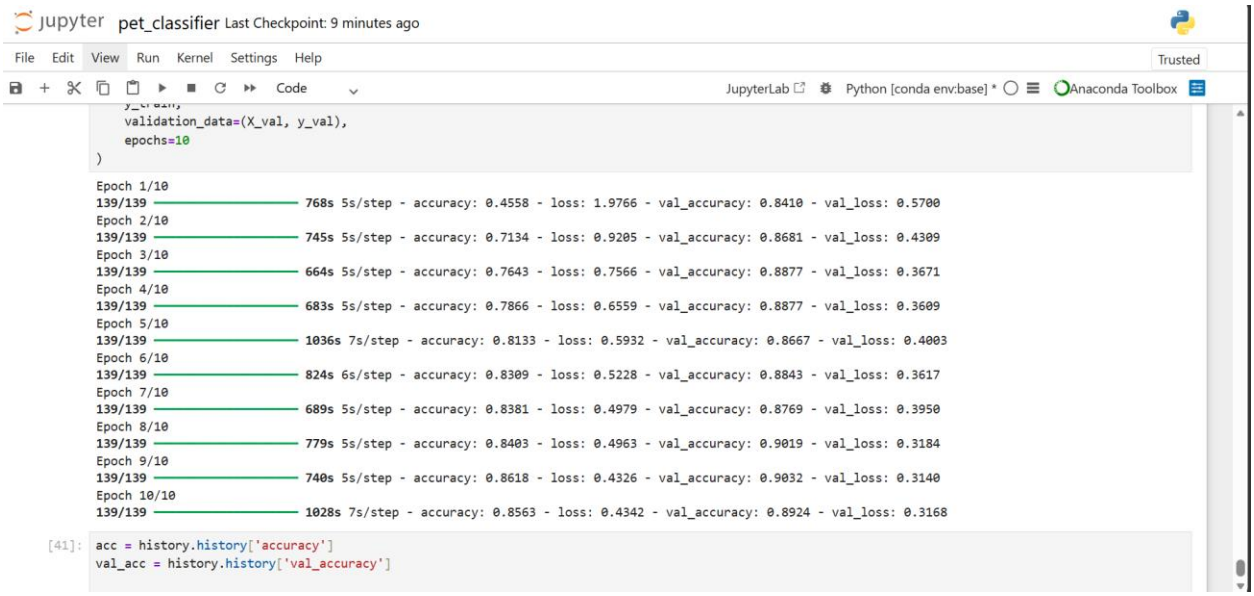
Code
JupyterLab
Python [conda env:base] *
Anaconda Toolbox
x = data_augmentation(inputs)
x = pp_i(x)
x = resnet_model(x, training=False)
x = Dropout(0.2)(x)
outputs = prediction_layer(x)
model = Model(inputs, outputs)
[39]: model.compile(optimizer=Adam(), loss=CategoricalCrossentropy(), metrics=['accuracy'])
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.
[40]: history = model.fit(
X_train,
y_train,
validation_data=(X_val, y_val),
epochs=10
)
Epoch 1/10
139/139 768s 5s/step - accuracy: 0.4558 - loss: 1.9766 - val_accuracy: 0.8410 - val_loss: 0.5700
Epoch 2/10
139/139 745s 5s/step - accuracy: 0.7134 - loss: 0.9205 - val_accuracy: 0.8681 - val_loss: 0.4309
Epoch 3/10
139/139 664s 5s/step - accuracy: 0.7643 - loss: 0.7566 - val_accuracy: 0.8877 - val_loss: 0.3671
Epoch 4/10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Code
JupyterLab
Python [conda env:base] *
Anaconda Toolbox
validation_data=(X_val, y_val),
epochs=10
)
Epoch 1/10
139/139 768s 5s/step - accuracy: 0.4558 - loss: 1.9766 - val_accuracy: 0.8410 - val_loss: 0.5700
Epoch 2/10
139/139 745s 5s/step - accuracy: 0.7134 - loss: 0.9205 - val_accuracy: 0.8681 - val_loss: 0.4309
Epoch 3/10
139/139 664s 5s/step - accuracy: 0.7643 - loss: 0.7566 - val_accuracy: 0.8877 - val_loss: 0.3671
Epoch 4/10
139/139 683s 5s/step - accuracy: 0.7866 - loss: 0.6559 - val_accuracy: 0.8877 - val_loss: 0.3609
Epoch 5/10
139/139 1036s 7s/step - accuracy: 0.8133 - loss: 0.5932 - val_accuracy: 0.8667 - val_loss: 0.4003
Epoch 6/10
139/139 824s 6s/step - accuracy: 0.8309 - loss: 0.5228 - val_accuracy: 0.8843 - val_loss: 0.3617
Epoch 7/10
139/139 689s 5s/step - accuracy: 0.8381 - loss: 0.4979 - val_accuracy: 0.8769 - val_loss: 0.3950
Epoch 8/10
139/139 779s 5s/step - accuracy: 0.8403 - loss: 0.4963 - val_accuracy: 0.9019 - val_loss: 0.3184
Epoch 9/10
139/139 740s 5s/step - accuracy: 0.8618 - loss: 0.4326 - val_accuracy: 0.9032 - val_loss: 0.3140
Epoch 10/10
139/139 1028s 7s/step - accuracy: 0.8563 - loss: 0.4342 - val_accuracy: 0.8924 - val_loss: 0.3168
[41]: acc = history.history['accuracy']
val_acc = history.history['val_accuracy']

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Code

JupyterLab Python [conda env:base] * Anaconda Toolbox

```
139/139 ━━━━━━━━━━━━━━━━━━━━ 1028s 7s/step - accuracy: 0.8563 - loss: 0.4342 - val_accuracy: 0.8924 - val_loss: 0.3168
```

[41]:

```
acc = history.history['accuracy']
val_acc = history.history['val_accuracy']

loss = history.history['loss']
val_loss = history.history['val_loss']

epochs_range = range(len(acc))

plt.figure(figsize=(15, 8))

plt.subplot(1, 2, 1)
plt.plot(epochs_range, acc, label='Training Accuracy')
plt.plot(epochs_range, val_acc, label='Validation Accuracy')
plt.legend(loc='lower right')
plt.title('Training and Validation Accuracy')

plt.subplot(1, 2, 2)
plt.plot(epochs_range, loss, label='Training Loss')
plt.plot(epochs_range, val_loss, label='Validation Loss')
plt.legend(loc='upper right')
plt.title('Training and Validation Loss')

plt.show()
```

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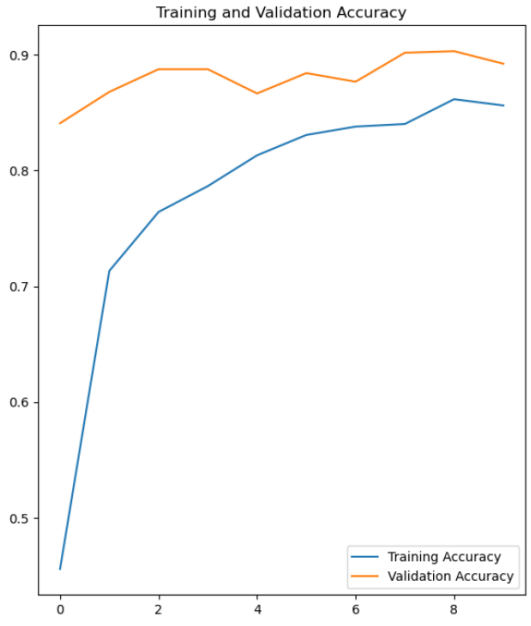
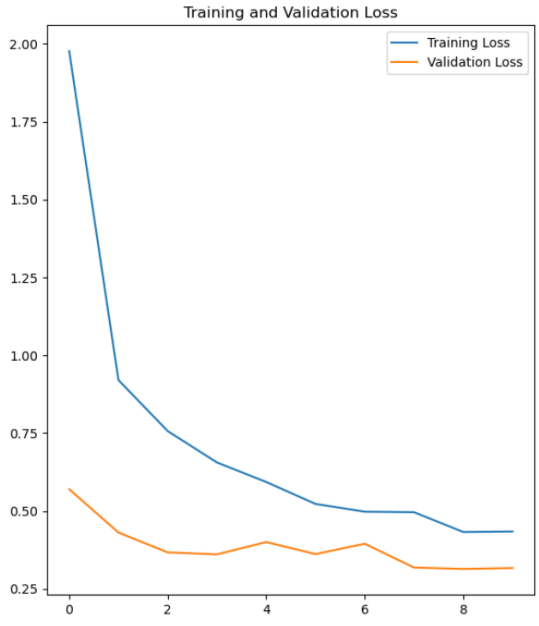
Code

JupyterLab Python [conda env:base] * Anaconda Toolbox





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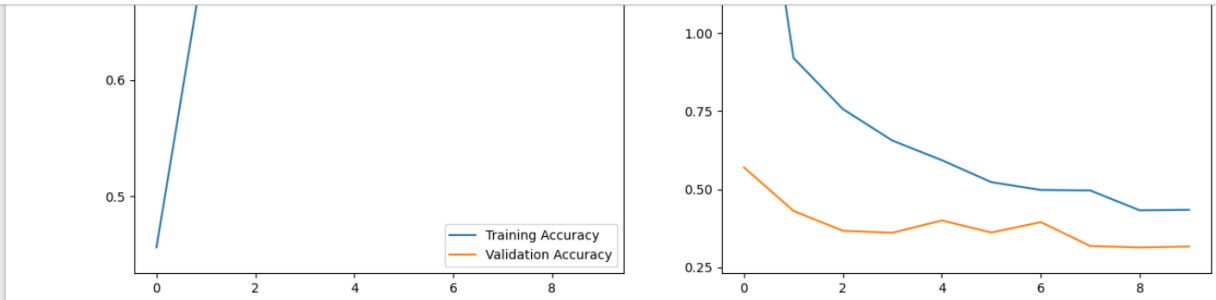
Code JupyterLab Python [conda env:base] * Anaconda Toolbox



[42]:
```
model.evaluate(X_test, y_test)
```

```
47/47 ━━━━━━━━━━━━━━━━━━━━ 279s 6s/step - accuracy: 0.8843 - loss: 0.3340
```

[42]:
```
[0.3340424597263336, 0.884303092956543]
```

[43]:
```
y_pred = model.predict(X_test)
```

```
47/47 ━━━━━━━━━━━━━━━━━━━━ 309s 6s/step
```

[ ]: